

# Cross-Constituency Aggregation for Community Notes

Emreca Ulu, Jingyao Shi, Carina I. Hausladen  
University of Konstanz

<https://github.com/vulonviing/cross-constituency-aggregation-community-notes>

## Abstract

Community Notes is X’s crowd-based fact-checking system: contributors write contextual notes, others rate them, and a bridging algorithm decides which become visible. That algorithm avoids simple majority capture by estimating note quality separately from ideological alignment, so a note should gain support across disagreement rather than within one side [30, 31]. But because participation is sparse and unequal, display decisions can turn on a small active core, and many notes stay in Needs More Ratings for want of diverse support [10, 24]. We argue this is not only a data-sparsity problem but an aggregation problem: cross-group agreement is treated indirectly, through a fitted model, rather than as an explicit decision rule over constituencies.

We take inspiration from political science and social choice, where systems such as Switzerland’s double-majority referendum treat cross-group support as part of a decision’s legitimacy. A note should become visible not because one active group overwhelms the rest, but when each relevant constituency has had a real chance to support it. We operationalize this as cross-constituency aggregation: recover behavioral constituencies from co-rating patterns among 200,000 raters, compute each note’s approval rate inside each, and combine those rates with a geometric mean under a minimum-coverage rule. In a 100,000-note slice, Community Notes displays 6,832 of 44,722 representative-picked notes (15%); our rule identifies a 13,655-note rescue pool, raising potential visibility to 20,405 notes (46%), of which a second-stage screen judges 8,558 sourced and rescue-worthy. Useful notes can remain hidden when cross-group support is modeled only indirectly.

## 1 Introduction

Community Notes is X’s crowdsourced fact-checking system. When a post may need context, contributors write a short note and others rate whether it is helpful; if it gathers enough support it is shown under the post. Rather than professional fact-checkers or classifiers, a crowd decides which context becomes visible [30, 31].

The key innovation is a bridging algorithm. A simple majority vote would be easy for one political side or one active group to capture, so X scores notes with a matrix factorization over the sparse note-rater matrix (Section 2), crediting a note only for support that viewpoint compatibility cannot explain. That factorization is the core ranking layer rather than the full

pipeline, which adds agreement safeguards, engagement and independence checks, and a November 2025 geometric-mean pilot.

The design is powerful but incomplete. Note helpfulness is still estimated from sparse, uneven ratings, which recent work shows is sensitive to noisy, strategic, and hyperactive raters [10, 20], and participation is steeply unequal (Section 3), so a formally open crowd can still turn on a small active core. More basically, the model keeps group support inside a fitted estimate: it never forms a quantity such as “approval inside constituency A” versus “inside constituency B,” so cross-group acceptance is present only indirectly.

We start from a different premise. Instead of estimating which raters are more reliable as individuals, we treat recovered behavioral constituencies as legitimate parts of the crowd and ask whether a note is acceptable across them. This turns Community Notes from a prediction problem into a crowd-agreement problem, since the platform is deciding, in the name of “the community,” whether a note should attach to a public post:

When ratings come from behaviorally distinct constituencies, which aggregation rule should decide whether a Community Note becomes visible?

The inspiration is political science and social choice rather than another rater-quality model. Divided societies use power sharing, double majorities, parallel consent, or veto-like rules so that collective decisions depend on support traveling across groups [11, 16, 19, 25], and social-choice theory gives a formal version of the same idea [21, 23]. We borrow this logic in a limited way: Community Notes is not a state and displaying a note is not passing a law, but both face the same aggregation problem under disagreement. It yields four criteria:

- **P1 (Presence).** Every constituency must be present in the decision.
- **P2 (Non-compensation).** One constituency’s rejection must not be fully offset by another’s enthusiasm.
- **P3 (Symmetry).** Constituencies enter the rule symmetrically, so no group is privileged by default.
- **P4 (Behavioral recovery).** With no constituencies given in advance, they must be recovered from rating behavior rather than assigned from outside labels.

We operationalize these criteria as cross-constituency aggregation. Using the active 200k Representative pipeline, we

cluster active raters from co-rating behavior, reassign them into two main behavioral constituencies, compute each note’s approval rate inside each constituency, and combine these rates with a geometric mean under a minimum-coverage requirement. The geometric mean creates a soft veto: if either constituency strongly rejects a note, the final score falls, even if the other constituency is highly supportive. We then use this score to identify hidden Community Notes that deserve a second look, and we validate those candidates separately for sourcing, content type, and rescue-worthiness.

On a 100,000-note slice of the live sample, the rule surfaces a 13,655-note rescue pool the platform currently leaves hidden, of which a second-stage screen keeps 8,558 as sourced and rescue-worthy. Community Notes therefore has an aggregation problem as well as a rating problem: useful public-interest notes can remain hidden when cross-group support is modeled only indirectly.

## 2 How Community Notes Works

Visibility is decided not by majority vote but by a matrix factorization the platform calls “bridging” [30], which decomposes the predicted rating  $\hat{r}_{un}$  that rater  $u$  gives note  $n$  as:

$$\hat{r}_{un} = \mu + i_u + i_n + f_u \cdot f_n \quad (1)$$

Four components isolate the note’s baseline helpfulness:

- $\mu$  (**global intercept**): the platform-wide baseline rating tendency.
- $i_u$  (**rater intercept**): whether a rater is generally generous or strict.
- $i_n$  (**note intercept**): the **target metric**, the note’s helpfulness stripped of ideological alignment and rater bias.
- $f_u \cdot f_n$  (**viewpoint match**): how well the note aligns with the rater’s political or behavioral ideology.

### A Critique of Matrix Normalization: Profiling Over Consensus

The terms  $i_u$  and  $f_u \cdot f_n$  do not measure “bad faith”; they act as normalization tools. But by relying on them, the core action of Community Notes shifts from evaluating a note’s *content* to estimating the user’s *intent*. Simulation studies of the open-source algorithm make this concrete: a coordinated minority of just 5–20% of biased raters can suppress genuinely helpful notes [29]. Observers have questioned the system’s narrow reliance on cross-group consensus [12]; our concern is more specific. To isolate note quality ( $i_n$ ), the algorithm treats rater tendencies and viewpoint compatibility as structural noise (Figure 1), and so profiles users and reads their intentions: is this user voting on objective truth, or cheering for their side? The result is an implicit hierarchy of raters, on the assumption that weighting “high-quality” evaluators is the most effective way to govern visibility.

That intent-reading framework has a fatal flaw. Goyal et al. [10] and Nudo et al. [20] show such quality-estimation structures to be highly vulnerable to strategic voters and “hyperac-

tive minorities”: inferring rater quality from a sparse matrix, the model leans on whoever rates most often, so a hyperactive minority’s voting patterns dictate the latent space, and an algorithm hunting for “quality raters” inside such a lopsided pool amplifies that core rather than capturing community consensus.

### The Algorithm in Practice: A Conceptual Map

Two of the algorithm’s cases are easy to state. A partisan rater who approves a note flattering their own side earns it nothing: the viewpoint match  $f_u \cdot f_n$  already predicted that vote, so  $i_n$  stays flat and the note stays hidden [30]. The reverse case is where the model earns its keep. When two raters at opposite poles independently mark the same note “Helpful,” viewpoint alignment can explain nothing, the surplus has nowhere to land but  $i_n$ , and the note surfaces as carrying something true enough to cross partisan lines [5].

The third case is corrosive rather than additive. Both stories above assume the algorithm holds an honest map of where each rater sits in the latent space. It does not start with one; it draws the map from the rating record, and a few hands draw most of it. Think of a class graded on participation where the teacher keeps asking “what does the class think?” but the same handful of students answer every session, until their voices become the teacher’s idea of the whole room. Community Notes inherits this distortion: a hyperactive, polarized minority supplies most ratings, so their behavior fixes where the “viewpoint” axis lies and which votes look “surprising” [20]. Once a narrow crowd has drawn the map, the “enemies” whose handshake looked so convincing may be two wings of one faction, and the bridge the model celebrates spans a gap most users were never placed on.

## 3 The Gap in Community Notes

### The Platform’s Defense: Bridging

X defends its algorithm with a “bridging” analogy [31]: a note supported from only one side of the polarization axis stays hidden, and visibility requires positive ratings from raters the model places at opposite ends of the latent viewpoint dimension. This, the argument runs, prevents majority tyranny and resists coordinated manipulation by any single faction.

### Why the Defense Does Not Hold

The bridging claim rests on a prior step that undermines it. Before any cross-group comparison can take place, the formula of Section 2 must situate every rater in a two-dimensional space of inferred identity. Two terms do this work.  $i_u$  collapses each rater to a leniency or strictness tendency, from which a rater who is critical because they apply high standards cannot be distinguished from one who systematically opposes certain content. The viewpoint term  $f_u \cdot f_n$  goes further, pinning each rater to a coordinate on an axis the model derives entirely from co-voting patterns; a “Helpful” rating counts as cross-group

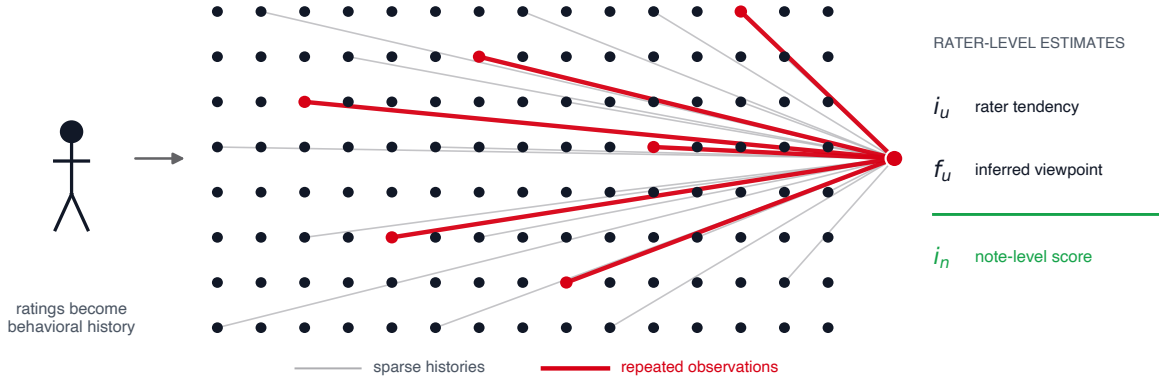


Figure 1: Repeated participation gives highly active raters more leverage over the fitted latent space: no explicit vote weight, just more observations behind the estimates of  $i_u$  and  $f_u$  that the pipeline uses before isolating  $i_n$ . Schematic.

| Scale             | $k$ | Initial spectral partition |         |          | After Method B |          | Hyperactive outlier            |
|-------------------|-----|----------------------------|---------|----------|----------------|----------|--------------------------------|
|                   |     | ARI                        | Pearson | Discrim. | Pearson        | Discrim. |                                |
| 60k               | 2   | 0.989                      | -0.69   | 86%      | —              | —        | absorbed (none)                |
| 150k              | 3   | 0.977                      | -0.65   | 83%      | -0.644         | 82.9%    | 62 users ( $\sim 7\times$ )    |
| 200k <sup>†</sup> | 3   | 0.971                      | -0.62   | 82%      | -0.620         | 81.4%    | 64 users ( $\sim 11\times$ )   |
| 210k              | 3   | 0.969                      | -0.62   | 81%      | -0.617         | 81.2%    | 66 users ( $\sim 12\times$ )   |
| 240k              | 4   | 0.955                      | -0.60   | 80%      | -0.596         | 80.0%    | 60 + 1,431 ( $\sim 25\times$ ) |
| 250k              | 2   | 0.997 <sup>‡</sup>         | —       | —        | —              | —        | degenerate: 249,933 / 67       |

Table 1: Clustering stability across scale. Initial columns: the partition selected on the raw co-rating graph before reassignment,  $k$  chosen by bootstrap stability, ARI the mean bootstrap ARI at that  $k$ . Method B columns: the final two-camp partition after vote-profile reassignment, on notes with at least ten raters per camp. Pearson correlates the camps’ note-level approval rates; Discrim. is the share of notes whose between-camp gap exceeds 0.3. The last column gives the separated sub-population and its median vote count relative to the main camps.

<sup>†</sup>Production setting.

<sup>‡</sup>Stability by itself certifies nothing. Eight of the 19 AMG variants on the ladder selected a  $k = 2$  that separates 55 to 70 power-voters from every other rater, at bootstrap ARI between 0.976 and 1.000; the 250k row is that failure mode at its clearest.

signal only when the model could not have predicted it from that coordinate. Bridging, operationally, means “this approval was surprising given where we placed this rater,” not “two independent communities both endorsed this note.”

The resulting action is intent inference rather than content evaluation. The algorithm asks not *what* a note says, but *who* tends to approve notes like this. Sparse, unequal participation makes the inference less reliable still: when a small hyperactive minority supplies most of the votes that fix the latent axis, the coordinate system reflects their behavioral patterns, not the broader community’s.

## Evidence from Our Own Pipeline

To test whether scale resolves the participation problem, we ran a clustering ladder of 19 AMG variants from 70k to 250k users, anchored by two ARPACK baselines at 50k and 60k (Table 1). Wherever the algorithm’s own selection produces two comparably sized camps, that structure is robust: bootstrap ARI above 0.95, between-camp correlation strongly negative. A hyperactive outlier reappears at every scale above 60k, power-voters 7 to 25 times more active than ordinary raters; larger samples do not dissolve this group, they rediscover it.

The ladder also shows why a stability score cannot be read as a certificate. Eight of the 19 variants scored bootstrap ARI between 0.976 and 1.000 for a partition that merely fences off roughly 60 power-voters, at 250k splitting 249,933 against 67 (Table 1): perfectly reproducible and analytically worthless.

Three scales survived that filter with two large camps and at most one small outlier cluster: 60k, 200k, and 210k. We take 200k as the production setting, on a joint reading of stability, camp balance, between-camp correlation and discriminating share rather than a maximization of any one; 210k reproduces the same structure at slightly weaker values.

Even at 200k the algorithm does not hand us two camps. Bootstrap stability selects  $k = 3$ , and the third cluster holds 64 raters with a median of 1,269 votes each, against 121 and 101 in the main camps. Leaving it in place would grant a 64-person clique the standing of two constituencies of roughly a hundred thousand raters; dropping it would discard the votes that most heavily shape the co-rating graph. We therefore reassign its members to the camp whose note-level voting they track, yielding the final split of 107,734 and 92,266 raters. The cost is small: the between-camp correlation and the discriminating share each move by less than a percentage point (Table 1), so absorbing the power-voters leaves the two-camp structure intact rather than manufacturing it.

## The Problem, Stated

Two failures sit beneath this gap, and only one is about data. The first is design: the algorithm profiles individuals instead of consulting communities, asking whether a rater’s approval was predictable rather than whether the groups that genuinely disagree have each consented. Sample size is irrelevant; with perfectly balanced participation the model would still read intent. The second is empirical, and our ladder demonstrates it: the latent axis on which the bridging check rests is fixed by a hyperactive minority, so even the “cross-group agreement” the algorithm rewards is agreement across a distorted pair of poles. Razuvayevskaya et al. [24] measure this skew from the outside, where the top 10% of contributors produce 58% of all notes, and our experiments confirm it from the inside. The gap is not a shortage of ratings. It is an aggregation rule that reads individual intent and then mistakes the most active raters’ consensus for the community’s.

## 4 Learning from Divided Societies

Section 3 left the production rule trying to bridge groups it cannot name. If cross-group acceptability is to be the display standard, the institutions built to decide across durable division are the natural place to look for an aggregation rule.

### The Display Decision and Legitimacy

Whether that gap matters depends on what the display decision is for. Each time the platform turns thousands of ratings into a single show/hide call, it does more than rank content: it attaches a public correction to someone’s post and presents it as coming from “the community.” Readers cannot opt out. A displayed note changes what everyone sees, including the people who think it is wrong, and its force comes from that shared environment rather than from any legal power [13]. The design question is therefore not which note is best but which rule should decide when the crowd disagrees — and a plain majority always returns an answer, though not always one the losing side has reason to accept.

Community Notes has already accepted the premise. The bridging criterion exists because the platform agrees that a note supported by one side alone should not appear, so cross-group acceptability is its display standard today, expressed as a latent estimate rather than as groups. Users respond to the difference: compared with a context-free misleading flag, a community-written note raises trust in fact-checking [7]. Asking about legitimacy introduces no foreign requirement; it restates a commitment the system already holds, in a form that can be checked.

What we take from states is the aggregation logic, not the political theory around it. Legitimacy here means something thin and procedural: whether the groups affected by a decision have reason to accept the way it was made.

| System                                                                                               | Cross-community requirement                                                                                                                                               |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Switzerland</b> | Double majority: federal referenda require approval both by a national popular majority and by a majority of cantons [17].                                                |
|  <b>Belgium</b>     | Linguistic parity in the federal cabinet; major legislation requires parallel consent from Dutch- and French-speaking majorities in parliament [16].                      |
|  <b>Bosnia</b>      | Vital-interest veto: any of the three constituent peoples (Bosniaks, Croats, Serbs) may block legislation deemed destructive to their community [4].                      |
|  <b>N. Ireland</b>  | Parallel consent or weighted majority required for key Assembly votes, so decisions cannot pass without support from both the unionist and nationalist designations [19]. |

Table 2: Examples of constitutional systems that require cross-community or cross-regional approval.

### Constitutional Design in Divided Societies

Democracies that govern divided societies repeatedly decline to resolve persistent disagreement by averaging over it; instead they institutionalize the cleavage structure itself and set the validity condition of a collective decision as concurrent acceptability across its component parts [16]. Table 2 summarizes four canonical cases: the Swiss double majority [17], Belgian parity between linguistic blocs [16], Bosnian cross-group veto rights [4], and Northern Irish parallel consent [19].

The rules differ, some territorial, some linguistic, some ethno-national, but share one logic: these democracies treat their cluster structure as politically meaningful in its own right. That logic is also the source of non-compensatoriness. A key vote of the Northern Ireland Assembly requires unionist and nationalist support at once, and support from one side cannot be traded for a victory on another bill. Acceptability must hold within that decision, with neither side dispensable — the structure a geometric-mean soft veto encodes rather than a weighted average. Power-sharing is not costless, and Section 9 returns to the limits of the borrowing.

The carry-over to crowdsourced moderation is therefore not to estimate what each constituency latently thinks, which would be quality recovery displaced from notes to clusters, but to honor the cluster structure itself, making each cluster’s acceptability an independent condition for the display decision.

### Proportional and Representative Social Choice

The constitutional record is qualitative; social-choice theory gives the same logic formally. A legitimate collective decision should stay responsive to the distinct constituencies rather than collapsing them into one pooled electorate [21, 22], and approval-based proportional representation makes the point axiomatically: large cohesive coalitions must not be washed out by majority aggregation [3, 26]. Most directly relevant is Qiu’s *representative social choice* [23], which targets exactly the setting where constituencies must be inferred from behavior rather than given ex ante — the platform’s situation, and the basis for the behavioral-recovery criterion.

Community Notes is, structurally, a repeated single-item acceptance setting, the object of the fair public decision-making literature [6, 15, 28]. Unlike that literature, which places fairness at the sequential level, so that a constituency outvoted on one issue can be compensated on later ones, the display decision is an unexitable individual act made in the community’s name. There is no sequence across which to amortize, which reinforces the non-compensatoriness already obtained from parallel consent.

Together the two literatures fix the validity condition: acceptability must be *concurrent* across constituencies rather than averaged over a pooled total, those constituencies must be inferred from behavior rather than given *ex ante*, and, because the display decision is unexitable and cannot be amortized across decisions, acceptability must hold *non-compensatorily* within a single note. The next section states these, with the symmetry requirement they imply, as four operational criteria.

## 5 Cross-Constituency Aggregation

### Design Criteria

The argument above narrows the design space to four requirements on any rule deciding under structured disagreement, each answering to one component of CCA. Behavioral clusters are not nations and we do not claim they hold such groups’ normative standing; these are *weakened* aggregation requirements adapted to platform governance, collected with their sources in Table 3.

**P1 (Presence).** No display decision may be made while any constituency is absent; missing representation must not be convertible into apparent consensus. (*Discharged by a presence floor of three raters per constituency, not a reliability threshold.*)

**P2 (Non-compensation).** One constituency’s strong rejection cannot be offset by another’s enthusiasm; acceptability must hold within each constituency, not merely in the cross-constituency average. (*Discharged by geometric-mean aggregation, a soft veto.*)

**P3 (Symmetry).** Constituencies enter the decision on equal footing, independent of size or rating activity. (*The geometric mean is symmetric in its arguments and takes approval rates, not raw counts.*)

**P4 (Behavioral recovery).** Since no *ex-ante* partition of raters exists, constituencies must be recovered from observed behavior transparently and reproducibly. (*The one criterion with no constitutional analogue. Discharged by spectral clustering on co-rating similarity, with stability diagnostics.*)

### Rule Definition

The rule is stated over two behavioral constituencies, recovered by the clustering of Section 6. Let  $a_{ic}$  be note  $i$ ’s approval rate inside constituency  $c$ , and  $H_{ic}$ ,  $N_{ic}$  the helpful and total

| Principle                     | Requirement                 | Source (§4)                  | CCA component        |
|-------------------------------|-----------------------------|------------------------------|----------------------|
| <b>P1</b> Presence            | No absent constituency      | Double majority (CH)         | Coverage rule        |
| <b>P2</b> Non-compensation    | No cross-bloc offsetting    | Mutual veto (BA, NI)         | Geometric mean       |
| <b>P3</b> Symmetry            | Equal footing, any size     | Parity rules (BE)            | Symmetric aggregator |
| <b>P4</b> Behavioral recovery | Blocs inferred, not labeled | Representative social choice | Spectral clustering  |

Table 3: The four design criteria at a glance: each principle, its requirement on the display decision, its source (Section 4), and the CCA component that discharges it.

ratings it receives there:

$$a_{ic} = \begin{cases} H_{ic}/N_{ic}, & N_{ic} > 0, \\ 0.5, & N_{ic} = 0. \end{cases} \quad (2)$$

The 0.5 fallback marks abstention, not endorsement. Presence (P1) forbids abstention from standing in for support, so both constituencies must contribute at least three raters before the note is scored at all. Three ratings do not measure approval precisely; they establish that the constituency was there. Community Notes applies minimums of the same shape in its own scoring [31], and set at three the requirement rarely bites (Section 7). For a note that clears the bar, the cross-constituency score is the geometric mean

$$C_i = \sqrt{\max(a_{i1}, \epsilon) \max(a_{i2}, \epsilon)}, \quad (3)$$

with  $\epsilon = 10^{-6}$  guarding against a hard zero. A note is CCA-selected when  $C_i > 0.5$  and the coverage requirement is met; when Community Notes does not already display it, the selected note enters the CCA rescue pool.

The geometric mean is the minimal aggregator that meets P2 and P3 at once: symmetric in its arguments, increasing in each on  $(0, 1]^2$ , and collapsing to zero as either rate does. A plain minimum would also veto, but it ignores the stronger side entirely; a weighted average would let one side’s surplus offset the other’s shortfall, which P2 forbids. The rule is thus the product of the four criteria rather than reverse-engineered to fit the data, though we do not claim it is the unique function with these properties.

### Objections and Scope

Read against these criteria, the production model’s gap is precise. Eq. (1) contains a counterpart to P2, since the bridging term credits helpfulness ( $i_n$ ) only for approval the viewpoint match cannot explain; but the constituency form of P2, that acceptability hold separately within each group, cannot be stated at all, because groups are not model objects. P1 has no counterpart either: pooled estimation has no notion of “a group being absent.” P3 is likewise absent, a larger bloc simply contributing more terms to the loss, and the polarization

safeguard conflicts with P2, discounting notes on *polarizing issues* rather than notes with *one-sided support*. The reliability-weighted extension [10] inherits the same structure. This is the design failure of Section 3: the rule reads individual intent where the criteria ask for community consent.

Our own rule invites objections in turn. The first concerns P4: by what right do algorithmically recovered clusters acquire the standing of constituencies? In constitutional regimes that standing comes from history, language, or identity; on the platform it can come only from the structure of the disagreement itself, and we require three things of a partition before it counts. It must *persist*: under resampling the algorithm keeps recovering two large camps beside a small hyperactive cluster at every non-degenerate scale (Table 1). That is nowhere near sufficient, as the ladder’s own failures show, since the variants that merely fence off a power-voter clique score highest of all on stability. Each side must therefore be *large* enough that cross-constituency agreement is a demand on two real publics rather than statistical noise, and ours each hold tens of thousands of raters; the 64 power-voters are not a third, since under a symmetric rule their veto would weigh as much as that of a camp a thousand times larger. And the two must *discriminate*, reaching different verdicts on a large share of notes; about 81% do. A partition meeting all three is not an arbitrary artifact but the lines along which acceptance and rejection run. The standing is conditional: when rating behavior no longer shows a stable cluster structure, CCA’s precondition fails and pooled estimation regains its warrant.

An immediate worry follows: is a two-class behavioral partition not just the model’s ideology axis split at the sign of  $f_u$ ? Even when both are binary, they are different objects. The  $f_u$  split requires a cut point on a single jointly fitted line; the behavioral split comes from the full co-rating similarity structure, and its boundary need not correspond to any threshold on that line. Topic-level diagnostics point away from coincidence, since neither constituency is uniformly more approving across issues. There is also a procedural reason to keep them apart:  $f_u$  is estimated by the very model under audit, and using the audited rule’s own instrument to delineate the parties would be improper.

A second worry concerns rescue itself: why should a note that fails the platform’s cross-ideological test regain value under cross-constituency consent? They are versions of one intuition tested with different instruments. X asks whether a note’s support is surprising given the viewpoint coordinates it infers; CCA asks whether each of two behaviorally recovered groups approves. The rescue pool is what that difference produces, and three things open it: the polarization safeguard can read “the issue divides people” as “the support is one-sided”; real blocs differ along several dimensions where the model gives each rater one coordinate, so bridging the axis and travelling across blocs come apart; and the latent estimate is strongly regularized, hence conservative by construction [10].

CCA therefore does not overturn the platform’s answer to a shared question; it marks the notes where the answer may come from how the standard is implemented rather than from

the standard itself. When rater disagreement is unstructured noise around a shared quality signal, pooled estimation is well warranted; it is the persistence of behavioral blocs (Section 3) that makes the cross-constituency question the right one here.

## 6 Approach

### 6.1 Data and Pipeline

The pipeline follows the repository’s active 200k Representative setup, with its thresholds in Appendix A. Figure 2 gives the filter chain: 1.7M English-language notes down to a dense 100k-note slice and the 200k raters active on it. The two floors of three, on candidate notes per tweet and ratings per note, secure a genuine multi-note choice set and minimum observed coverage (Appendix A.1); the 48-hour window keeps ratings tied to the note’s active review period. The slice spans 44,722 distinct tweets, the unit from which the Representative strategy selects one note. The dense core is deliberate: co-rating similarity is informative only when users rate enough of the same notes to be compared.

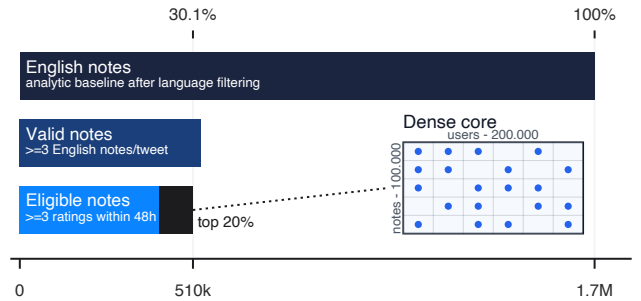


Figure 2: Construction of the analytical sample: 1.7M English notes, filtered to tweets with at least three English candidates and notes with at least three eligible ratings within 48 hours, yielding 510k; from these a 100k-note dense slice and its 200k active raters.

### 6.2 Recovering Constituencies

Section 5 states behavioral recovery (P4) as a criterion; this is how we discharge it. The 200k scale balances coverage against diagnostic quality: as less-active raters entered the sample, shared-note overlap thinned while a small hyperactive core remained among the graph’s most reproducible structures [20, 24]. Table 1 reports the ladder and Appendix A.2 the selection argument.

On that slice we build a sparse, mean-centered user-by-note matrix, each entry a rater’s deviation from their own average rating, and embed a symmetric nearest-neighbor co-rating graph spectrally (Appendix A.3). Constituencies are therefore recovered from rating behavior alone, independently of any quantity X’s model estimates. Evaluating  $k$  from 2 to 7, we select the one with the highest mean bootstrap ARI: at 200k that returns  $k = 3$ , at ARI 0.971 against 0.593 for  $k = 2$ , while



silhouette points the other way. The criteria diverge because a direct  $k = 2$  specification must represent two structures with only two labels, the division between the large camps and a recurring clique of power-voters.

Section 3 describes how the 64 power-voters of that third cluster are reassigned by vote profile (Method B) to reach the final split. Nothing downstream turns on how that is done: an embedding-centroid variant (Method A) sorts 29 of the 64 differently, and both partitions return the same between-camp correlation of  $-0.620$  and the same 81.4% discriminating share (Appendix A.4).

### 6.3 Selection

For each tweet, the Representative strategy selects the candidate note with the highest cross-constituency score (Eq. (3)), defined only once both constituencies have contributed at least three ratings (Section 6.1) — the coverage rule that operationalizes P1. A note is CCA-selected when its bridge score exceeds 0.5, and newly rescued when it is CCA-selected but not already shown.

### 6.4 Validation

Aggregation and content quality are different judgments: CCA identifies notes clearing a cross-constituency support bar, and an independent stage asks whether the note is worth showing. We use Gemma 4 31B IT, the strongest open-weight model that runs unquantized on our two-GPU allocation (Appendix B.1), judging each of the 13,655 notes in the frozen rescue pool through a single-turn, zero-shot, rubric-based prompt (Appendices B and C). Rubric-based LLM-as-a-judge measurement has broader empirical precedent [2], though none validating this model or rubric (Appendix B.2).

The model sees only the note’s own text: not the tweet it responds to, no cited URL, and not its upstream selection score. This is a real constraint on what the stage can certify (Section 9), since the judgment concerns the note as a self-contained piece of writing, not the truth of its claim or its relevance to the post.

Validation begins with a content screen and ends with scoring. Stage 1 sorts each note into one of five content categories, requiring a visible external source pointer before it can count as sourced factual information; unsourced, spam and hostile notes stop here. Opinion- or speculation-labeled notes get a second reading, Stage 1.5, asking whether a substantial sourced factual core survives once the subjective framing is set aside; those that pass join the direct Stage 1 passes in a common Stage 2 pool. Stage 2 scores every note there for source traceability, the strength of the link between claim and source, clarity and neutrality of language, and constructive rather than hostile presentation, passing at 50 or higher. Throughout, a missing source is scored as a validation failure, not as evidence that the claim is false: the rubric measures what the note visibly demonstrates about itself.

Model revision, prompt text, input manifest, decoding seed and all 25,804 attempt statuses are recorded, so any judgment

| Note set                                   | Count  |
|--------------------------------------------|--------|
| Representative-picked note universe        | 44,722 |
| CN shown within representative picks       | 6,832  |
| CCA-qualified                              | 20,405 |
| shown by CN                                | 6,750  |
| hidden (rescue pool)                       | 13,655 |
| Below CCA threshold                        | 24,317 |
| Rescue pool: Gemma-validated ( $\geq 50$ ) | 8,558  |
| Rescue pool: below score 50                | 1,818  |
| Rescue pool: stopped at content screen     | 3,279  |

Table 4: Representative picks shown by Community Notes versus the CCA rescue pool and its Gemma validation outcome.

can be regenerated (Appendix B.3); one Stage 1 note remained unresolved.

## 7 Results

### 7.1 Rescue Pool and Visibility

Of 44,722 representative-picked notes, Community Notes currently displays 6,832, about 15%. The CCA rule qualifies 20,405 of the same universe, 46%. The two sets are not nested: 6,750 CCA-qualified notes are already shown, while 82 shown notes fail the CCA threshold outright, so CCA disagrees with the platform in both directions. Setting those 82 aside, the 13,655 CCA-qualified notes Community Notes does not display form the rescue pool the rest of this section evaluates (Table 4, Figure 3). Coverage is rarely why the remaining picks fall short: only 71 never receive three ratings from both constituencies, and across the full 100k-note slice only 185 notes go unscored for lack of coverage. The rest are scored and fall short on the merits, at a median bridge score of 0.291.

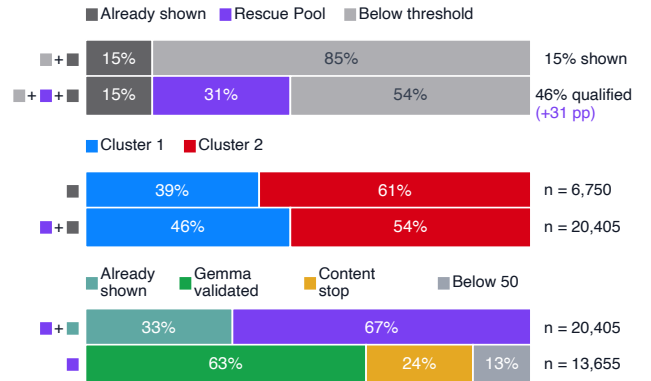


Figure 3: Among 44,722 representative-picked notes, 20,405 meet the CCA threshold: 6,750 are already shown and 13,655 form the hidden rescue pool, of which Gemma validates 8,558. 3,279 stop at the content screen and 1,818 score below 50.

## 7.2 Validation Outcomes

Stage 1 sorts the 13,655-note rescue pool by content type: 10,096 notes carry a visible external source and read as sourced factual information, while 1,703 read as opinion or speculation, 1,340 as irrelevant or spam, 373 as unsourced claims, and 142 as hostile in tone. Stage 1.5 recovers 280 of the opinion-labeled notes, giving a Stage 2 population of 10,376. Of these, 8,558 clear the 50-point threshold, 62.7% of the original pool, while 1,818 score below it and 3,279 stopped at the content screen. Only 31 of the validated notes came through Stage 1.5; the other 8,527 passed Stage 1 directly.

Figure 4 plots both stages together: most of what the content screen stops is opinion without a sourced core, and most of what survives scores well above the display threshold rather than barely over it, with 43% in the top 90-to-100 band.

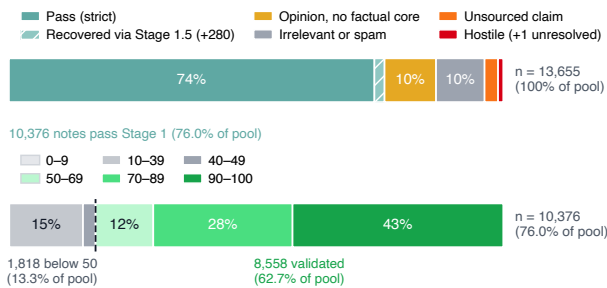


Figure 4: Two-stage Gemma validation of the 13,655-note rescue pool. Upper bar: the Stage 1 content screen with the Stage 1.5 recall of 280 opinion-labeled notes carrying a sourced factual core. Lower bar: the 10,376 surviving notes scored 0–100 against the 50-point threshold.

## 7.3 Topical Structure

Section 5 argued from the clustering ladder alone that the two constituencies are not simply a strict camp and a lenient one. Grouping 2,181 topically labeled notes into 13 subject areas and computing each constituency’s average approval gives that claim a direct test (Figure 5): Cluster 2 approves more readily on 8 of the 13 topics and Cluster 1 on the other five, including Israel/Palestine content, Iranian politics, and the Gaza hostages. A single strictness axis would predict one constituency leading everywhere by a roughly constant margin; the topic breakdown shows disagreement that is topic-specific instead, consistent with distinct viewpoints rather than one careful reader and one permissive one.

## 7.4 Illustrative Rescue Cases

Aggregate counts conceal two distinct ways CCA changes a display decision. We therefore inspect six purposively selected cases (Tables 5 and 6): three from the 8,339 Gemma-validated selections on tweets where Community Notes shows nothing, and three from the 219 where the tweet already has a different

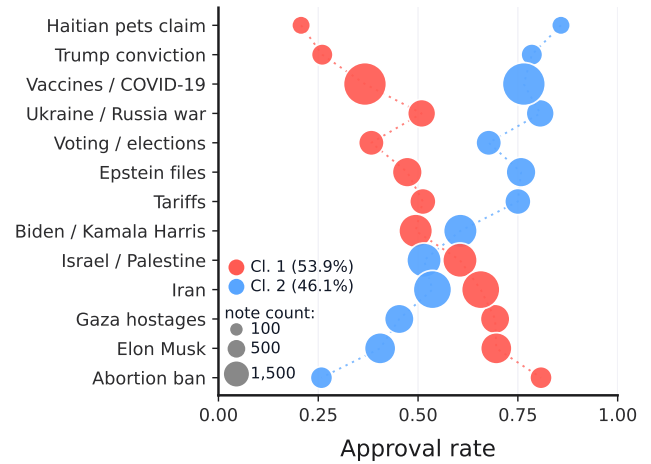


Figure 5: Average approval rate by constituency across 13 subject areas; bubble size encodes note count. Leadership flips between constituencies, and the gap ranges from 0.090 on Israel/Palestine content to 0.651 on a fabricated claim about Haitians and pets.

displayed note. They illustrate the rule’s behavior rather than estimate either population’s average quality.

The hidden cases show that rescue is not confined to one type of controversy: a public-health correction, a diplomatic-history note, and a ballot-eligibility note, each citing a primary source, each on a tweet that displays nothing.

The paired comparisons make the qualitative gain concrete. The displayed Yarmouk note describes Mu’tah, whereas the CCA alternative identifies the battle depicted and distinguishes the two events; in the Anne Frank case the alternative separates a removed graphic adaptation from the original diary instead of treating them as one title; the last pair corrects the pictured person’s identity before explaining Viganò’s status. CCA raises the bridge score in all three. These purposively selected examples support a comparative judgment about directness and contextual fit, not a ground-truth estimate for the full pool.

## 8 Discussion

Section 4 argued that a display decision made in a collective name owes the affected parties a procedure they have reason to accept, and that Community Notes already concedes cross-group acceptability as its standard without being able to state it directly. The results give that argument empirical content: P1 through P4 are no longer only design commitments but each attached to a number. Presence rarely has to act as a hard veto, itself evidence the requirement is working quietly, since almost every note that reaches a bridge score reaches it having been heard from both sides. Non-compensation is visible in the Yarmouk comparison, where the CCA alternative has slightly lower raw approval but stronger cross-constituency support, and so a higher bridge score. Symmetry holds regardless of which constituency is larger or more active. Behavioral recovery is the clustering ladder of Section 3, which holds from



| Case                         | CCA-selected note (verbatim)                                                                                                                                                  | Global | CCA score | Gemma |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------|-------|
| Raw chicken safety           | Raw chicken does not need to be washed before cooking. Washing poultry [...] increase bacterial contamination in the kitchen. <i>[CDC; USDA]</i>                              | 95.1%  | 0.941     | 98    |
| Eisenhower and Suez          | Factual error: U.S. President Dwight D. Eisenhower insisted that Israel withdraw from Arab territory during the 1956 Suez Crisis. <i>[Eisenhower's message to Ben-Gurion]</i> | 84.6%  | 0.817     | 98    |
| Trump–Colorado ballot ruling | The Supreme Court unanimously ruled Trump could not be taken off the ballot by Colorado [...] because states do not have the power to do so. <i>[the Court's opinion]</i>     | 92.1%  | 0.735     | 98    |

Table 5: Validated CCA selections for three tweets on which Community Notes displays no note. Quotations are abridged with [...] and sources named in brackets rather than printed; full text and URLs are in the repository. “Global” is overall approval, “CCA score” the geometric mean of the two constituency approval rates, “Gemma” the Stage 2 score.

| Case                          | Decision | Note (verbatim)                                                                                                                                                                                                              | Global | CCA score | Gemma |
|-------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------|-------|
| Yarmouk/Mu'tah                | X shown  | The Battle of Mu'tah was fought between the Byzantine Empire and the First Islamic State. [...] It ended with a Byzantine victory. <i>[Wikipedia; islam365]</i>                                                              | 92.2%  | 0.774     | —     |
|                               | CCA      | The post refers to the Battle of Yarmouk (636 AD), but reliable estimates put Muslim forces at 15,000–40,000, not 3,000 [...]. 3,000 Muslims fought at the earlier Battle of Mu'tah (629 AD). <i>[Wikipedia; Britannica]</i> | 91.2%  | 0.837     | 95    |
| Anne Frank graphic adaptation | X shown  | Diary of a Young Girl, by Anne Frank, is part of the 8th-grade curriculum of the Florida Department of Education. <i>[FL DoE standards]</i>                                                                                  | 97.8%  | 0.781     | —     |
|                               | CCA      | Florida removed the graphic novel adaptation of Anne Frank's diary due to nudity deemed unsuitable for kids, but the original diary remains available. <i>[WAFB]</i>                                                         | 97.8%  | 0.848     | 95    |
| Cardinal Burke/Viganò         | X shown  | Mr Viganò is a former archbishop who was excommunicated from the church in 2024 for schism. <i>[The Pillar]</i>                                                                                                              | 89.9%  | 0.783     | —     |
|                               | CCA      | The image shows Cardinal Raymond Burke, not Archbishop Viganò. Viganò was excommunicated for schism in July 2024 and holds no Church office. <i>[Vatican News; Catholic Herald]</i>                                          | 94.2%  | 0.871     | 95    |

Table 6: Same-tweet cases where CCA selects a stronger alternative to the displayed note. Each pair holds the tweet’s only displayed note and the higher-scoring CCA selection; quotations are abridged as in Table 5. Columns as above, with “Gemma” reported only for the CCA-selected notes.

60k to 240k raters at every non-degenerate scale. Because CCA is derived from stated criteria, its design can be evaluated one commitment at a time against the evidence.

One development is visible in the platform’s own direction. In November 2025 X began piloting a final-scoring model that aggregates helpfulness through a geometric mean across the rater-factor spectrum [31]: the soft-veto logic of CCA, adopted in production but still organized along one fitted axis. The step CCA formalizes is the obvious next one, taking that geometric mean over behaviorally recovered constituencies, each required to be present, rather than over points on an estimated ideology line.

CCA is not the only response to the profiling critique of Section 2. Quality-sensitive matrix factorization [10] keeps that structure but weights each rater by an estimated reliability, reaching comparable accuracy from 26–40% fewer ratings. The two do not compete on the same axis: QSMF asks which raters are more reliable, CCA which constituencies approve, and a platform could run both, with reliability weighting inside each constituency and cross-constituency consent as the gate on top. That combination is future work, not a claim we test here.

Language-model judgment stays downstream of selection here: CCA forms the rescue pool and Gemma evaluates its contents afterward. A future system could fold LLM judgment into selection itself, as X’s For You stack already places Grok-derived transformers inside retrieval, ranking and policy

classification [32], though we neither specify nor test such an integration.

The rescue pool’s size speaks to a question broader than Community Notes. Some of the gap is a rating-volume problem, the sparse participation Section 2 describes. But the topical pattern in Figure 5, where leadership flips between constituencies and the gap varies severalfold across topics, is not explained by volume alone; it points to an aggregation rule that reads uneven cross-group support as absence of consensus rather than consensus expressed unevenly. Crowdsourced moderation that wants to credit genuine cross-group agreement needs a rule that can see the groups, not only a model that infers them while estimating something else. Given how much more trust a community-written note carries than a bare misleading flag (Section 4), which notes earn that framing is not a minor implementation detail.

## 9 Limitations

The empirical exercise evaluates one fully specified implementation, and several limits accompany the design.

The constitutional experience in Section 4 clarifies a trade-off rather than supplying a template. Power-sharing can protect a group from domination while hardening group boundaries, empowering actors who claim to speak for a bloc, and creating veto points that frustrate broadly supported decisions

[18, 19, 27]. CCA faces a platform-scale version: treating two behavioral constituencies as the units of consent could preserve a division that might otherwise weaken, and the soft veto could block a note supported by a broad but not overwhelming majority. The rule therefore applies only while co-rating behavior supports stable, substantively sized constituencies; if that structure disappears, forcing a group-based decision would no longer satisfy P4 and pooled estimation becomes the better baseline.

Our final partition is a simplification, and one we impose in part. Folding the third cluster back into the main camps is defensible on the grounds given in Section 3, but it is a decision about standing rather than a finding about the data, and political disagreement on the platform is not thereby shown to be binary. Behavioral constituencies need not track any external category such as party or country; we make no claim that they do, and a finer partition might recover distinct behavior our two-camp setup collapses together.

Content validation carries a different set of boundaries. Gemma judges each note as a self-contained piece of writing: it never sees the tweet the note responds to or opens a cited source, so a passing score certifies visible sourcing and internal coherence, not that the source supports the claim or that the note addresses the post. Giving the model the tweet and the cited material would broaden the task to include those judgments, but we tested no retrieval-enabled setup, no other model, and no sensitivity sweep over the 0.5 bridge and 50-point thresholds. Nor is content validation immune to the manipulation risk that motivated Section 2: an aggregation rule that resists a hyperactive rating minority still depends on a note-quality check never stress-tested against adversarial writing [29].

Coverage is bounded on several fronts at once. The dataset is English-only, restricted to a 48-hour rating window, and drawn from roughly the top 20% of notes by rating volume, so the results describe the densest part of the platform, not its long tail. The presence floor that discharges P1 leaves thinly rated notes unscored rather than risk a decision on too little evidence; we set it at three ratings per constituency and did not test higher floors, which would cut sampling noise at the cost of leaving more notes unscored. Broader participation could make some excluded notes scorable, but would not change the verdict on the tens of thousands that already have coverage and fall below the bridge threshold. The reported configuration is a fully documented implementation, not evidence that these settings are optimal.

## 10 Conclusion

Community Notes already commits to cross-group acceptability as its display standard; it just cannot state that standard directly, because its rating model never forms the constituencies the standard is about. This paper recovers them from rating behavior and makes the standard explicit: a geometric mean of within-constituency approval, gated by a coverage requirement that keeps any constituency from being counted without being

heard. Applied to the platform’s own representative picks, the rule surfaces 13,655 notes that clear cross-constituency support and remain hidden, of which 8,558 also pass an independent content check. Neither number is the platform’s final answer on any of these notes; both are evidence that a meaningful share of what disappears into Needs More Ratings does so for a reason the aggregation rule creates. The claim is not specific to Community Notes: any system deciding visibility, in a collective name, over a durably divided audience faces the same choice (Section 4) between compressing disagreement into one pooled score and preserving its structure long enough for every relevant part of the audience to be heard from. Cross-constituency aggregation should be treated as an alternative design feature of crowdsourced moderation systems.

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## A Pipeline and Constituency Recovery

This appendix holds the operational detail behind Section 6. Every threshold and variant reported here is also implemented in the accompanying repository.

### A.1 Filter thresholds and the two three-rater floors

A floor of three applies at both the candidate-note and the rating stage, but for distinct operational reasons: the first guarantees a genuine multi-note choice set, while the second follows the same minimum-coverage logic that X applies before cross-viewpoint evidence can affect a note’s status. The rating floor establishes minimal observed participation rather than claiming a reliable estimate of approval [31].

### A.2 Selecting the 200k rater scale

The scale was chosen by testing rather than assumed. The tests encountered the same participation imbalance identified for Community Notes in Section 3: as progressively less-active raters entered the sample, shared-note overlap thinned, while a small hyperactive core remained among the graph’s most reproducible structures. We selected 200k as the best balance between coverage and diagnostic quality. 210k reproduced the same two-camp structure somewhat more weakly, while larger configurations introduced additional activity-driven outliers or collapsed into a power-voter-versus-rest split.

### A.3 Spectral embedding and outlier handling

A symmetric 15-nearest-neighbor graph over the mean-centered user-by-note matrix defines co-rating similarity, and a spectral embedding, computed with algebraic multigrid to keep the procedure tractable at this scale, places each rater in a low-dimensional space shaped by whom they tend to agree with. Candidate values of  $k$  from 2 to 7 are evaluated by silhouette score and by stability under repeated resampling. Where a direct  $k = 2$  specification is forced, it sometimes recovers the balanced camp split across resamples and sometimes uses one label to isolate the small, highly active group; the same high-activity pattern reappears at neighboring scales. Clusters smaller than  $\max(1\% \text{ of raters}, 500)$  are treated as outliers rather than granted constituency standing.

### A.4 Reassignment: Method A versus Method B

Method B is not another  $k = 2$  spectral fit. It begins from the more stable  $k = 3$  partition, keeps the membership of both large camps fixed, and classifies only the 64 outliers, correlating each outlier’s note-level vote profile against the two main constituencies’ aggregate profiles. Method A, an embedding-centroid variant, sorts the same 64 raters by distance in the spectral embedding instead. The two rules disagree about 29 of the 64, who together cast 0.07% of the ratings in the slice, and the resulting partitions return the same between-camp correlation of  $-0.620$  and the same 81.4% discriminating share. Method B is the canonical partition used throughout the paper; the original spectral label survives only as a diagnostic column.

## B Gemma Validation Runtime

Table 7 records the canonical configuration of the validation run.

### B.1 Model selection and serving constraints

Gemma 4 31B IT was the strongest feasible open-weight model for this role. In the April 2, 2026 Arena snapshot, its thinking variant scored 1,452 and ranked third among open models, while its 30.7-billion-parameter dense BF16 weights could run without quantization on our two-GPU allocation [1, 8].

### B.2 Rubric-based LLM-as-a-judge precedent

Across 1,141 human-labeled binary classification tasks, GABRIEL reports a median accuracy of 0.85 for GPT-5 and finds measurements generally robust to semantically equivalent prompt variants [2]. This benchmark supports the feasibility of large-scale qualitative judgment, but it does not validate Gemma or the task-specific rubric used here.



| Component            | Canonical configuration                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model                | google/gemma-4-31B-it; 30.7B dense parameters; revision 518276fb130dc81caf9a4f772e65e63ef2526493; instruction-tuned thinking variant; BF16; no quantization.                                                                                                                                                               |
| Hardware             | Two NVIDIA L40S GPUs, 46,068 MiB reported memory each; tensor parallelism 2. SCCKN AMD EPYC host with eight allocated CPU slots; SGE request of 64 GB h_vmem per slot and approximately 52 GB observed peak maxvmem (host memory, not GPU memory).                                                                         |
| Serving              | Locally hosted vLLM 0.25.0 inference server, bound to 127.0.0.1 and accessed through its /v1 HTTP endpoint; CUDA 13.2; PyTorch 2.11.0+cu130; Transformers 5.13.1; asynchronous scheduling; prefix caching; 64 concurrent requests and maximum sequences; 32,768 maximum batched tokens; 90% GPU-memory-utilization target. |
| Decoding             | Thinking enabled with the Gemma 4 reasoning parser; temperature 0.2; 8,192-token context cap; 4,096-token completion cap; deterministic seed from SHA-256 of base seed 42, stage, note ID, and attempt number.                                                                                                             |
| Prompt contract      | One note per independent user prompt; no system prompt, few-shot examples, guided generation, tweet context, URL fetching, retrieval, embeddings, vector store, upstream selection score, or Gabriel label. The model receives only the note text and stage-specific instructions.                                         |
| Validation and retry | Unconstrained generation followed by strict JSON output validation; maximum three attempts. Every attempt retains its seed and status; returned outputs are stored with the parsed result, reasoning, and token counts, while response failures retain error metadata.                                                     |
| Run scale            | 25,734 logical judgments across Stages 1, 1.5, and 2; 25,733 accepted judgments from 25,804 attempts; 99.75% valid on the first attempt; 32,113,128 prompt-and-completion tokens; 12h 28m summed active-phase time and 15h 51m from first to final output, including inter-stage gaps.                                     |

Table 7: Canonical Gemma validation configuration. Model-family properties and the Arena snapshot come from the official model materials and Arena leaderboard [1, 8, 9]; run-specific values come from the frozen manifests, SCCKN scheduler records, worker configuration, and per-attempt audit logs; vLLM is described by Kwon et al. [14].

### B.3 Audit trail and output validation

Every returned response is stored with its parsed result, reasoning, and token counts; the 71 response errors retain their seed and error metadata but contain no output fields because no model response was returned. Generation is not schema-constrained. Responses are parsed and checked against a strict contract afterward: an exact key set, a label drawn from the fixed list or an integer score in range, and a reason under forty words, with at most three attempts before a judgment is left unresolved. This audit trail allows any individual judgment to be regenerated independently while preserving the distinction between model output and post-generation validation.

## C Canonical Gemma Validation Prompts

The three prompts are not reproduced here. They are published verbatim, with the SHA-256 hashes below, at `data/llm_validation/PROMPTS.md` in the repository, alongside the tracked artifacts the canonical run consumed. At inference time `{note_text}` was replaced by the text of one Community Note; no system prompt or additional contextual message was supplied. Stage 1 classifies a note into one of five content categories, Stage 1.5 rechecks opinion-labeled notes for a sourced factual core, and Stage 2 scores rescue-worthiness. The hashes were computed over the template text before the tracked artifact’s terminal newline:

Stage 1: fbbd4bd93fb419ad66cce097bb18a8705fd331f38ec1a9bf433aa783281023d2

Stage 1.5: 26114bb8632357166cf4cee9a4f0ce356ed1e9e527477d56c842e53415bc930f

Stage 2: 8c98c54b9c413ee70c161f40ec8e89f0b19ac6420b2bfe669e7ee1f9b136c644